

Research Paper: The Theory of Forced Verification Networks for Probabilistic Fraud Detection

A Novel Framework for Near-Zero Cost Probabilistic Fraud Detection in Large-Scale Financial Systems

1. Abstract

We present a novel mathematical framework for automated financial auditing systems, designed to ensure the integrity of massive datasets with near-zero computational overhead. The core principle is **Forced Error Propagation**, where transactions are interconnected within a directed acyclic graph (DAG) exhibiting strong expander graph properties.

We mathematically prove that any localized tampering in a single transaction propagates corruption through a chain of hashes across the network, achieving near-100% detection probability via inspection of only a very small random sample of size $O(\log N)$.

2. Core Mathematical Model

2.1 System Definition

Let $\mathcal{S}$ be a financial system containing a set of transactions

$$V = \{v_1, v_2, \dots, v_n\}.$$

The hash of each transaction v_i is defined to depend not only on its own data but also on the hashes of its neighbors $N(i)$:

$$H(v_i) = \mathcal{H}(\text{data}(v_i) \parallel \bigoplus_{j \in N(i)} H(v_j))$$

where:

- $\mathcal{H}$ is a cryptographic hash function (e.g., SHA-256)
- $\parallel$ denotes concatenation
- $\oplus$ denotes XOR (or any suitable aggregation)
- $N(i)$ is the set of predecessor transactions linked to v_i

2.2 Error Propagation Dynamics

If an adversary modifies transaction v_t to v'_t , then immediately:

$$H(v'_t) \neq H(v_t).$$

Any subsequent transaction v_k that includes v_t in its neighborhood ($t \in N(k)$) will have its stored hash invalidated:

$$H(v_k) = \mathcal{H}(\dots \parallel H(v'_t) \parallel \dots) \neq H_{\text{stored}}(v_k)$$

This triggers a **domino effect**. When the graph is an expander, the number of affected nodes after d propagation steps grows exponentially:

$$|I_d| \geq \min(n, (1 + \epsilon)^d), \quad \epsilon > 0$$

where ϵ is the expansion constant. Thus, contamination spreads extremely rapidly across the network.

3. Probabilistic Detection Theorem

3.1 Detection Probability

We aim to detect fraud by checking only a small random sample $S \subset V$ with $|S| = m$.

If the error has propagated to affect a fraction $\rho = \frac{|\text{Impact}|}{n}$ of the network, the probability that the entire sample is clean (detection failure) is:

$$P(\text{Fail}) = (1 - \rho)^m$$

Therefore, the probability of successful detection is:

$$P(\text{Detect}) = 1 - (1 - \rho)_m$$

3.2 Exponential Power of the System

In a well-constructed expander network, a single tampering event typically contaminates roughly half the graph ($\rho \approx 0.5$). The detection probability then simplifies to:

$$P(\text{Detect}) = 1 - \left(\frac{1}{2}\right)^m$$

Numerical examples for small m :

- $m = 10 \rightarrow P(\text{Detect}) \approx 99.90\%$
- $m = 20 \rightarrow P(\text{Detect}) \approx 99.9999\%$

Conclusion: Verifying only **20 transactions** provides near-absolute certainty of integrity — even in a database of billions of records.

4. Proposed Construction Algorithm: 3D-Linkage

To enforce the desired expander properties, we introduce the **3D-Linkage** algorithm for every new transaction T_i :

1. Temporal Link

Connect to the immediately previous transaction:

$$L1 = H(T_{i-1})$$

2. Account Link

Connect to the last transaction of the same account/user:

$$L2 = H(T_{\text{user_last}})$$

3. Algebraic Random Link

Connect to a deterministically chosen distant transaction (ensuring long-range jumps):

$$k = (i \times \text{Prime_Const}) \bmod n, L3 = H(T_k)$$

Final hash:

$$H(T_i) = \mathcal{H}(\text{Data}(T_i) \parallel L_1 \parallel L_2 \parallel L_3)$$

5. Conclusion

This framework reduces financial auditing from a linear $O(n)$ full scan to a constant-time probabilistic check $O(1)$ in terms of samples required. Any attempt to alter even one transaction forces recomputation of thousands of dependent hashes, rendering fraud computationally prohibitive and economically irrational.

The system achieves **provably high detection probability** with minimal overhead, making it suitable for real-time integrity verification in massive financial datasets.